To What ENDS:

Behavioral Responses to State-Level Vaping Flavour Bans

Shaoyi Yu

72146228

Abstract

This paper examines the impact of state-level e-cigarette flavor bans on vaping and other substance use behaviors among individuals under 30, using BRFSS data from 2016 to 2023. A difference-in-differences approach reveals that flavor bans significantly reduce current vaping, but this effect weakens among individuals reporting frequent poor mental health, suggesting reduced responsiveness among psychologically vulnerable users. Additionally, flavor bans are associated with increased use of combustible cigarettes and chewing tobacco, indicating possible behavioral substitution. These findings underscore the importance of accounting for mental health heterogeneity in public health regulation and highlight the unintended consequences that can arise from uniform policy interventions.

1 Introduction

Over the past decade, the rapid rise of ENDS (Electronic Nicotine Delivery Systems), commonly known as e-cigarettes, has transformed the public health landscape around the globe. Marketed as a safer alternative to combustible cigarettes, e-cigarettes have become especially

popular among adolescents and young adults. Studies show that ENDS use among youth has surged dramatically in the US, with over 2.5 million middle and high school students reporting use in 2022 alone (Park-Lee et al., 2022). While some view vaping as a potential harm reduction tool for smokers, growing concerns over youth initiation and long-term health effects have prompted policymakers to intervene.

A major contributor to e-cigarettes' popularity lies in their appealing flavors such as fruit, mint, and sweets. Ambrose et al. (2015) reported that 81% of youth e-cigarette users pointed to the availability of flavors as a prime reason for use. These flavors not only mask the harshness of nicotine but also create the impression of a safer, more recreational activity—contributing to youth experimentation and, for some, long-term use. As a countermeasure, one of the most prominent regulatory tools has been the flavor ban, implemented at various levels of government in the United States. These policies prohibit the sale of non-tobacco flavored e-cigarettes with the goal of reducing the appeal of nicotine, particularly among adolescents. Despite mass adoption of this ban across states, the effectiveness and unintended consequences of such policies have not been extensively studied and remain an ongoing debate. Do flavor bans reduce overall nicotine use, or do they simply shift consumers toward other substances—including traditional cigarettes, chewing tobacco, alcohol, marijuana, or unflavored vape products?

This paper investigates the causal impact of e-cigarette flavor bans on vaping behavior and explores potential spillover effects on other substance use. Specifically, I replicate and extend the findings of Cotti et al. (2024), who conducted a multi-state difference-in-differences analysis of flavor ban effects using BRFSS data. While the original study focuses on average treatment effects, I contribute new evidence by incorporating event study dynamics and subgroup heterogeneity. By bringing additional variation into the empirical strategy, my analysis not only validates existing findings with newly available data but also tests for robustness and deeper behavioral responses that may be masked in average effects.

In particular, I examine whether differences in mental health status generate nuanced

impacts of flavor bans. The link between mental health and substance use behavior has drawn increasing attention in recent years, in large part due to the COVID-19 pandemic, which worsened psychological distress and social isolation for many individuals. An expanding body of evidence suggests that individuals with poor mental health may be more likely to initiate or rely on nicotine products as a form of self-regulation or coping. As such, it is reasonable to expect that such users may respond more strongly to policy interventions that restrict access to their preferred nicotine products. Understanding how these vulnerable subgroups respond to regulatory changes is critical for designing more equitable and effective public health policies.

To investigate these questions, I employ a difference-in-differences design with two-way fixed effects (state and year) and individual-level controls to estimate the causal effects of state-level flavor bans on current vaping behavior. There exists a statistically significant negative impact on vaping usage due to the flavor ban, and it varies by mental health level. I also explore potential behavioral spillovers by evaluating whether such bans unintentionally increase usage of other substances. Though the main spillover effects are generally null, I conduct exploratory subgroup analyses to test whether responses differ. Notably, among those reporting frequent poor mental health, I find evidence of stronger substitution toward combustible tobacco. While these findings are not uniformly robust across all specifications, they highlight the importance of targeted policy design and raise important questions for future research on behavioral substitution and mental health vulnerability.

This study contributes to the literature in three key ways. First, it replicates and extends recent findings on the effects of flavor bans by incorporating newly available public data, which provides an updated assessment of treatment effects. Second, it sheds light on behavioral heterogeneity by mental health status, an aspect that has received limited attention in past policy evaluations. Third, it examines the possibility of unintended substance use spillovers, offering insight into the broader behavioral consequences of nicotine regulation. Altogether, these contributions help inform ongoing debates surrounding substance

use, public health equity, and regulatory designs.

2 Literature Review

Recent research has increasingly focused on the impact of e-cigarette flavor bans. Among these studies, Cotti et al. (2024) provide one of the most comprehensive empirical assessments. Using data from the Youth Risk Behavior Survey (YRBS) and Behavioral Risk Factor Surveillance System (BRFSS), they estimate the effects of flavor bans implemented by states and large cities between 2015 and 2021.

Employing a two-way fixed effect difference-in differences method with staggered treatment timing, the authors find that flavor bans are associated with notable reductions in frequent and everyday ENDS use among youth and young adults. On the other hand, there is little evidence to support the policy's effect on the extensive margin. If narrowing sample down to adults aged 18-21, there appears to be significant decrease in ENDS use, regardless of margin. The study also uncovers evidence of behavioral substitution. Following the implementation of a flavor ban, young adults become more likely to smoke combustible cigarettes. This substitution is considerable among the under-21 population and raises concerns about potential unintended consequences of otherwise well-intentioned regulation. As for other substances, there is no indication of spillovers.

The authors also noted a few limitations for the paper. Firstly, neither YRBS nor BRFSS data records difference in flavored and unflavored ENDS products. This restricts the analysis to detect only net effects on ENDS usage. Second, most statewide or citywide flavor bans happened during the onset of COVID-19. Though the study applied controls such as cumulative covid death counts per 100,000 people, it is highly possible that unmeasured effects of the pandemic has been captured. It is important to note that, these limitations are equally relevant to my studies. Another caveat is that, given the novelty of ENDS flavor restrictions, the estimated effects should only be interpreted as short-run results with limited observa-

tions. However, At the time of this writing, BRFSS has released 2 extra years of data. By incorporating both years into the analysis, this study partially addresses the issue and offers a broader perspective than prior work.

In this paper, I replicate and extend the findings of Cotti et al. using publicly available BRFSS data with additional years, introducing heterogeneity by mental health and exploring dynamic treatment effects. This allows for a deeper understanding of how regulatory policies such as flavor bans affect not just the average user, but also vulnerable subpopulations whose behavior may differ in both magnitude and direction.

3 Data

3.1 Overview of BRFSS

Our primary dataset is the Behavioral Risk Factor Surveillance System (BRFSS), collected by the CDC. BRFSS is the largest continuously conducted representative health survey system in the United States. It is collected annually through cellphone and landline interviews, gathering detailed information on health-related risk behaviors, chronic health conditions, and preventive service use from more than 400,000 adult participants. An intricate system of survey weights is applied to ensure representativeness at both the state and national level (Centers for Disease Control and Prevention (CDC), 2023a).

3.2 Sample Selection

This analysis uses BRFSS data from 2016 to 2023, excluding 2018, during which e-cigarette questions were not included. The sample is restricted to adult respondents residing in states with identifiable vaping policy environments and complete information on key outcome and control variables. Since ENDS usage is most prevalent among younger populations and these groups are disproportionately affected by flavor restrictions, the analysis further limits the sample to individuals under 30 years old to improve model precision and enhance the

relevance of estimates to the most impacted demographic.

3.3 Outcome Variables

The primary outcome variable is `ECIGNOW2`, which measures respondents' current use of ENDS at the time of the survey. Although it does not record exact usage frequency in days, the variable classifies individuals into three categories: everyday users, some-days users, and non-users. Recoding this variable allows for granular analysis of usage intensity. For example, I construct a binary indicator equal to one for everyday ENDS users and zero for all others, enabling targeted estimation of policy effects on the most regular users.

To investigate potential spillover effects, I also use four additional outcome variables:

- `USENOW3`: Current chewing tobacco use.
- `_RFBING6`: Binge drinking status.
- `_RFSMOK3`: Combustible cigarette use.
- `MARIJAN1`: Marijuana use.

Each outcome is recoded into a binary indicator equal to one if the respondent reports current use and zero otherwise.

3.4 Treatment Variable

The key treatment variable, `flavor_ban`, is a binary indicator constructed based on the implementation date of state-level flavor bans. If a respondent's BRFSS interview date occurs after the effective date of the flavor ban in their state of residence, the variable is coded as one; otherwise, it is zero. Assuming immediate policy effect, no lagged implementation is modeled.

3.5 Subgroup: Mental Health

To assess subgroup heterogeneity, I use the variable `MENTHLTH` from BRFSS, which records the number of days in the past 30 days a respondent experienced poor mental health, including stress, depression, or emotional problems. Based on response distribution, individuals are categorized into:

- **Good Mental Health**: 0 days of poor mental health.
- **Moderate Mental Health**: 1–9 days.
- **Poor Mental Health**: 10 or more days.

By interacting these mental health groups with the flavor ban treatment, I assess whether regulatory responses differ across psychological well-being.

3.6 Policy Controls

The following policy indicators are included to account for state-level regulations that may affect nicotine and substance use behavior:

- **Tobacco 21** (`t21`): Whether the state raised the legal tobacco purchase age from 18 to 21.
- **ENDS Taxation** (`ecig_tax`): Binary variable indicating whether the state taxed e-cigarettes at the time of survey. Due to variation in tax structure—for instance, Kansas charges $0.05 per mL, Minnesota applies a 95% wholesale tax, and Pennsylvania taxes both devices and liquids at 40%—this variable is coded as binary (**?**).
- **Combustible Cigarette Tax** (`tax`): Measured in dollars per pack (Centers for Disease Control and Prevention (CDC), 2023b).
- **Recreational Marijuana Legalization** (`rec_marijuana`): Indicates whether cannabis use was legal for recreational purposes at survey time.

- **Indoor ENDS Ban** (`post_ecig_indoor_ban`): Whether ENDS use was banned in indoor public places.

3.7 Socioeconomic Controls

To capture broader socioeconomic and health context, I include the following continuous variables:

- **Unemployment Rate** (`unemp`): Monthly statewide unemployment rate in percent (U.S. Bureau of Labor Statistics, 2023).
- **Monthly COVID-19 Death Rate** (`Death`): Monthly state-level COVID-19 death rate (National Center for Health Statistics, 2023). Monthly rates better reflect the contemporaneous public health burden than cumulative figures, which only increase over time.
- **COVID-19 Period** (`covid`): A binary dummy equal to 1 for responses between March 1, 2020, and December 31, 2022. Though year fixed effects are included, they do not account for within-year variation in pandemic timing and severity. This dummy helps isolate behavioral disruptions tied specifically to COVID-19.

3.8 Weights

The BRFSS provides a complex sampling weight variable, `LLCPWT`, designed to ensure population-level representativeness at the state level. In this study, all regressions are estimated using `LLCPWT` as analytic weights. Due to variation in missingness across different outcome variables and subsamples, each regression drops a different set of observations, which makes re-normalizing the weights infeasible in a consistent way across specifications.

To assess whether this affects our results, I conduct sensitivity checks using alternative weight normalizations and subsample restrictions. The core findings remain robust across

these variations, suggesting that the main results are not driven by the exact scaling of the weights. Therefore, the `LLCPWT` variable is applied without further adjustment throughout the analysis.

4 Empirical Strategy

4.1 Baseline Model

To estimate the causal effect of state-level e-cigarette flavor bans on current vaping behavior, I begin with a difference-in-differences framework using both Ordinary Least Squares (OLS) and logistic regression (logit), depending on the binary nature of the outcome variable. The baseline specification is as follows:

$$Y_{ist} = \alpha + \beta \cdot \text{FlavorBan}_{st} + \gamma X_{ist} + \lambda_s + \delta_t + \varepsilon_{ist} \tag{1}$$

Here, Y_{ist} represents the outcome of interest for individual i in state s and year t, such as current ENDS use or other substance usage. The variable FlavorBan_{st} is a binary treatment indicator equal to one if the respondent is surveyed after a flavor ban is implemented in their state. X_{ist} denotes a vector of individual-level and policy-level controls. State fixed effects (λ_s) and year fixed effects (δ_t) are included to absorb time-invariant state characteristics and common temporal shocks, respectively.

4.2 Mental Health Interaction Specification

In some specifications, I explore potential heterogeneity in treatment effects by mental health status. I include interaction terms between the flavor ban indicator and mental health subgroups to estimate differential effects:

$$
\begin{aligned}
Y_{ist} = \alpha + \beta_1 \cdot \text{FlavorBan}_{st} + \beta_2 \cdot \text{MHGroup}_i \\
+ \beta_3 \cdot (\text{FlavorBan}_{st} \times \text{MHGroup}_i) + \gamma X_{ist} + \lambda_s + \delta_t + \varepsilon_{ist}
\end{aligned} \tag{2}
$$

Where MHGroup_i is a categorical indicator for mental health status (e.g., Good, Moderate, Poor), and the interaction term FlavorBan $\times$ MHGroup captures the differential treatment effects by subgroup.

4.3 Robustness Check: Event Study

To analyze the dynamic effects of flavor bans over time, I estimate an event study model following the standard approach in policy evaluation (e.g. Goodman-Bacon (2021), Cotti et al. (2024)). The specification replaces the single treatment indicator with a set of leads and lags relative to policy implementation:

$$
Y_{ist} = \alpha + \sum_{k \neq -1} \beta_k \cdot D_{s,t+k} + \gamma X_{ist} + \lambda_s + \delta_t + \varepsilon_{ist} \tag{3}
$$

Here, $D_{s,t+k}$ is a set of dummy variables indicating k years before or after the flavor ban takes effect in state s, excluding $k = -1$ as the reference period. This formulation allows for visual inspection of pre-trends and heterogeneous post-treatment dynamics.

4.4 Robustness Check: Double Selection Lasso

To address potential omitted variable bias in high-dimensional settings, I employ the double selection Lasso approach developed by Belloni et al. (2014). This method involves selecting control variables in a two-stage process using L1-regularization and then estimating treatment effects in the final model. The steps are as follows:

1. Run Lasso regression of the outcome Y on all controls and the treatment variable.

2. Run Lasso regression of the treatment variable on all controls.

3. Take the union of selected variables from both steps and include them in the final OLS regression.

This approach improves covariate selection and prevents overfitting, ensuring valid post-selection inference in finite samples.

5 Results

5.1 Main Effect

Variable	**Coefficient**	**Std. Error**	**P-Value**
Flavor Ban	-0.0162	0.006	0.004
Poor Mental Health Days	0.0035	0.000	0.000
Age	-0.0020	0.000	0.000
Post Indoor ENDS Ban	0.0120	0.005	0.018
Tobacco 21	0.0224	0.006	0.000
ENDS Tax	0.0021	0.021	0.919
Combustible Cigarette Tax	-0.0018	0.003	0.544
COVID Dummy	0.0222	0.020	0.276
Monthly COVID Deaths	-0.00000179	0.00000287	0.532
Unemployment Rate	-0.0011	0.001	0.422

Table 1: WLS regression estimating the effect of ENDS flavor bans on current vaping behavior among individuals under 30. The dependent variable is a binary indicator for current e-cigarette use. The model controls for demographics, economic and tobacco policy covariates, as well as state and year fixed effects. BRFSS analytic weights are applied, and standard errors are clustered at the state level.

The weighted least squares results indicate that state-level flavor bans are associated with a statistically significant 1.6 percentage point reduction in the probability of current vaping behavior among individuals under 30 (coef = -0.016, p = 0.004). Poor mental health days are positively associated with vaping, with 1 extra day of mental distress increasing vaping probability by 0.3 percentage points. This suggests higher usage among individuals experiencing psychological distress and underscores the importance of examining heterogeneous

effects through subgroup analysis, which will be covered in the next subsection. Age has a small but significant negative effect on vaping, reflecting a decline in usage with increasing age within the sampled range.

Several policy variables also exhibit notable patterns. Interestingly, states with indoor ENDS bans are associated with higher vaping prevalence. While this may reflect partial behavioral displacement or increased user awareness, it is also plausible that the direction of causality runs the other way. Jurisdictions with already high vaping rates may have adopted stricter indoor restrictions in response to rising use. However, this interpretation should be considered with caution, as the timing of policy implementation relative to behavioral trends is complex. The Tobacco 21 law shows a positive and significant coefficient, potentially reflecting enforcement lags or substitution among underage users in states where compliance was weak. Taxes on ENDS and combustible cigarettes, as well as COVID-related controls, do not appear to have significant effects on vaping behavior in this specification.

Overall, the results closely align with the original findings reported by Cotti et al. (2024), reinforcing the credibility of their core conclusions. The negative and statistically significant association between flavor bans and current ENDS use is similar to their reported average treatment effects, despite differences in sample years, model adjustments (e.g. different age range), and weighting procedures. The direction and magnitude of the coefficients are consistent with a policy-induced reduction in vaping, particularly among younger adults.

5.2 Spillover Effect

Outcome Variable	**Variable**	**Coefficient**	**p-value**	**Significant** ($p < 0.05$)
Current Chewing Tobacco Use	flavor_ban	0.0117	0.036	Yes
Current Smoking	flavor_ban	0.0280	0.004	Yes
Binge Drinking	flavor_ban	-0.0264	0.116	No
Marijuana Use	flavor_ban	-0.0299	0.530	No

Table 2: Coefficients from WLS regressions estimating the spillover effects of ENDS flavor bans on other substance use among individuals under 30.

Table 2 presents estimates of the impact of state-level flavor bans on a variety of substance use behaviors. Among the four outcomes analyzed, statistically significant effects (at the 5 % level) are observed for current smoking and chewing tobacco use. Specifically, flavor bans are associated with a 2.8 percentage point increase in the likelihood of current smoking ($p = 0.004$) and a 1.2 percentage point increase in the probability of chewing tobacco use ($p = 0.036$). These results suggest potential unintended spillover effects. Restrictions on flavored ENDS may lead some users to substitute toward traditional combustible or smokeless tobacco products. In contrast, the coefficients for binge drinking and marijuana use are negative but not statistically significant, indicating no clear behavioral response in those domains.

These findings broadly replicate the results reported in Cotti et al. (2024), who also identified a positive spillover to cigarette use following e-cigarette restrictions. The alignment further strengthens the case that flavor bans can induce compensatory increases in more harmful nicotine consumption patterns. The convergence across independent specifications and updated data for both main results and spillovers not only validates the robustness of prior research but also demonstrates the external replicability of their findings, which is critical for informing evidence-based tobacco regulation. Notably, while the original study focused primarily on smoking, our analysis extends the literature by discovering statistically significant spillover to chewing tobacco use, highlighting multilateral behavioral shifts that may inform more comprehensive and policy design. The OLS regression results for chewing tobacco and combustible cigarettes are presented in Appendix table A2 and A3, respectively.

5.3 Subgroup Analysis

5.3.1 Vaping Behavior

To further explore heterogeneity in behavioral responses, we examine the different effect of flavor bans on vaping behavior by mental health status. Following previously established empirical strategy, we interact the policy indicator with mental health groupings based on the number of poor mental health days reported in the past month.

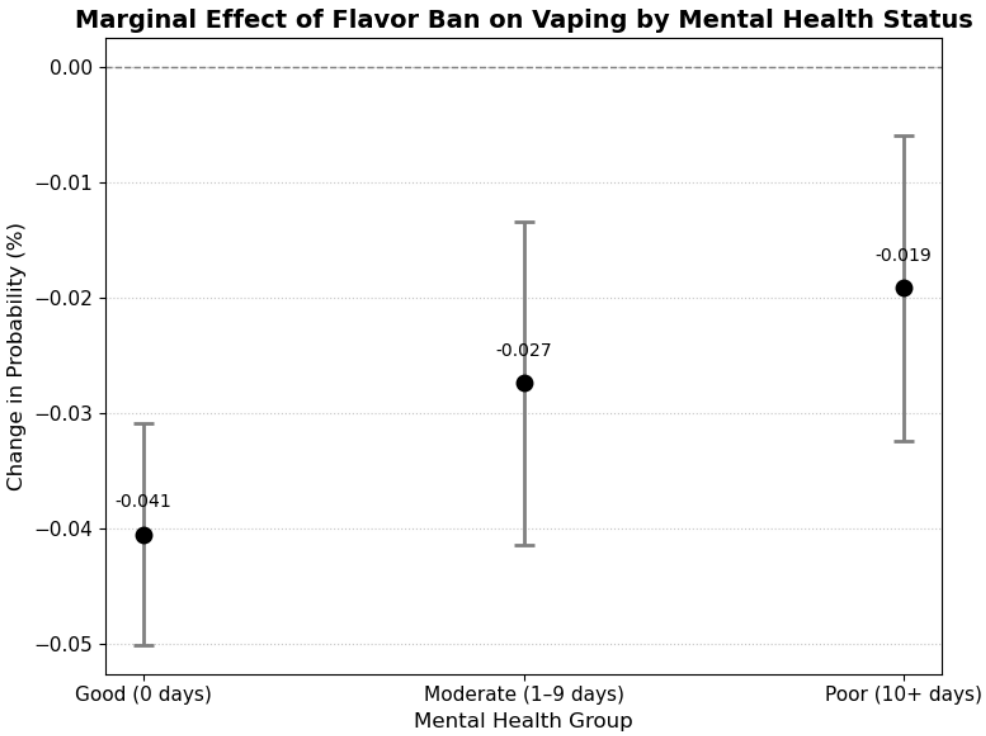


Figure 1: Average marginal effects of state-level flavor bans on current vaping behavior, stratified by self-reported mental health status. Results are based on a logit specification with interaction terms between treatment and mental health group (Good, Moderate, Poor). Diminishing magnitude of negative effects across the mental health status suggests reduced policy responsiveness among psychologically distressed individuals. Confidence intervals reflect 95% robust standard errors.

Figure 1 visualizes the estimated marginal effects of state-level flavor bans on current vaping behavior, stratified by self-reported mental health status. Individuals reporting good mental health exhibit the largest predicted reduction in vaping likelihood (–4.1 pp), followed by those with moderate (–2.7 pp) and frequent (–1.9 pp) mental distress. Although the confidence intervals overlap, the visible upwards trend suggests diminishing policy responsiveness among those experiencing greater psychological strain. This finding is consequential, as it suggests differential receptiveness to policy signals across mental health conditions. Vulnerable groups may exhibit behavioral inertia in response to regulatory changes, especially when the targeted behaviors serve as coping mechanisms. Such patterns highlight the challenges of influencing health-related behavior through uniform policy interventions and underscore

the need for tailored strategies in populations facing psychological distress.

While interaction terms in the linear probability model do not reach statistical significance, corresponding logit estimates suggest meaningful effect heterogeneity, particularly among individuals reporting frequent poor mental health. These divergent results underscore the importance of employing nonlinear specifications when investigating subgroup dynamics, as average marginal effects from OLS may obscure sharp shifts in behavioral probability concentrated within specific populations.

In contrast, estimates from the linear probability model using the same interaction specification are shown in Appendix Table A4. While the direction of the effects remains consistent, they do not generate statistically significant coefficients for any mental health subgroup. The lack of significance in OLS highlights a methodological dissonance between linear and nonlinear models. Importantly, it suggests that average treatment effects estimated via OLS may mask meaningful heterogeneity that becomes evident under a logit specification. We will further elaborate on the implications of this divergence in the Discussion section.

5.3.2 Combustible Cigarette Spillover

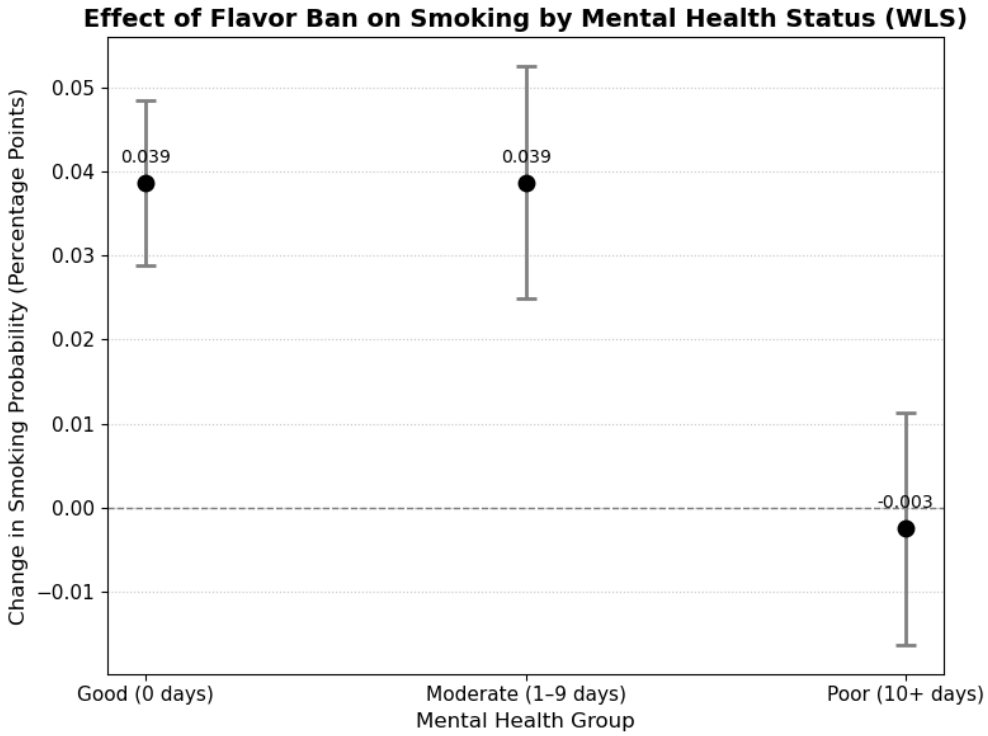


Figure 2: Estimated effects of flavor bans on combustible cigarette use by mental health subgroup using a linear probability model (OLS). Flavor bans are associated with increased smoking among those with good or moderate mental health, but a statistically significant decrease among individuals reporting frequent poor mental health. These patterns suggest limited substitution to smoking among more vulnerable users, potentially due to behavioral stickiness to ENDS products.

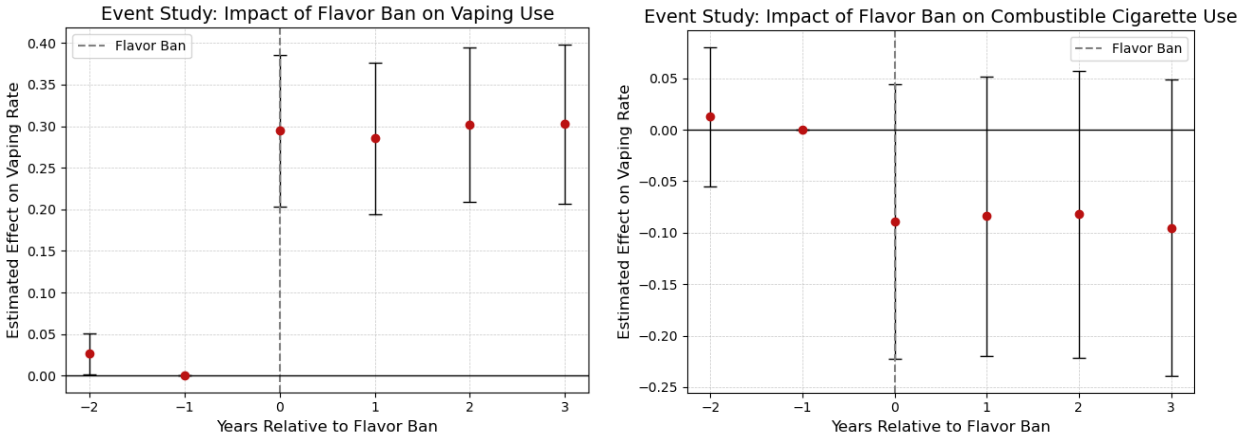
Figure 2 presents the OLS-estimated effect of flavor bans on combustible cigarette use across mental health subgroups. We find that the flavor ban is associated with a 3.9 percentage point increase in smoking prevalence among individuals reporting good mental health, and this effect is nearly identical for those with moderate mental health symptoms. However, among individuals experiencing frequent poor mental health, the effect is decreased and reversed in direction, indicating a small but statistically significant decrease in smoking. While the difference between the good and moderate groups is negligible, the trend suggests that individuals with worse mental health are less likely to switch to smoking in response to ENDS flavor restrictions. This forms a coherent story when compared to the subgroup

analysis for vaping. When policy changes occur, individuals with mental distress are very reluctant to give up ENDS, which explains the decrease in smoking.

Quite the opposite from the first subsection, this interaction is only detected in the OLS specification and corresponding logit models did not produce significant effects for any mental health group. One explanation is that the substitution behavior captured here operates more linearly across individuals and is better picked up by OLS than by the nonlinear logit framework, which may underweight marginal shifts at higher base probabilities. We further explore this divergence between models in the discussion section.

5.4 Robustness Check

5.4.1 Event-Study



(a) Event study: Main effect

(b) Event study: Spillover effect

Figure 3: Event study estimates of the effect of ENDS flavor bans on vaping use (a) and current cigarette use(b) among individuals under 30. In panel (a), we observe a delayed and positive response in cigarette use, whereas in panel (b), there is an immediate and persistent decline in vaping.

In Figure 3(a), we display the event-study estimates of the impact of flavor bans on vaping rates. The pre-ban years (-2 and -1) show effects approximately centered around zero, indicating no significant pre-trends before implementation and showing strong support to the parallel trends assumption. The post-ban years (0 to 3) shows a stark upward shift

in vaping, which conflicts with the linear model results.

In Figure 3(b), we plot the event-study estimates for combustible cigarette use. Unlike the vaping results, the pre-ban coefficients do not cluster around zero as neatly; the confidence intervals are wide, and the point estimates exhibit more variation, suggesting less stable pre-trends. Moreover, the post-ban coefficients move in the opposite direction of what we expect. Though with a wide confidence interval, smoking rates appear to decline instead of rise, and the sign flips relative to vaping behavior.

This counterintuitive directions in the panels is likely driven by limitations in the design: our study spans only a short window and includes a limited subset of states with implemented flavor bans This creates sparse time variation, with few periods before or after the policy change and smaller state-level sample sizes. Therefore, the event-study estimates may be biased or unstable. Even though we generated promising results for our OLS and logit model, such constraints inevitably reduce our confidence in the causal interpretation, and we will expand on these limitations and potential alternative explanations in the Discussion section.

5.4.2 Double Selection Lasso

To check the robustness of our model's covariate selection, we examine which variables were retained by the post-double selection Lasso procedure. Despite starting with a comprehensive set of controls—including demographic, policy, and temporal variables—only eight variables were selected: mental health days (`MENTHLTH`), age (`_AGE80`), sex (`SEXVAR_2`), several race indicators (`_RACE_2.0`, `_RACE_4.0`, `_RACE_8.0`), survey year 2020, and the state cigarette tax (Appendix Table A5). The chosen covariates are seen as predictive of either treatment assignment or the outcome.

The fact that many variables (e.g. state fixed effects, unemployment, COVID-19 indicators, other year dummies) were not selected indicates that they did not add sufficient predictive value for treatment or outcome beyond what was already captured by the selected subset. This suggests that some of the covariates we originally included for completeness may

have been superfluous or only weakly correlated with either dimension of confounding. In contrast, the inclusion of key demographic indicators and policy-related variables aligns with theoretical expectations and provides proof that the model has retained the most influential predictors.

Importantly, the consistency of the treatment effect estimate across the full model and the double Lasso specification supports the idea that our main results are not subject to overfitting or multicollinearity and appears robust to a more conservative, data-driven selection strategy. While double Lasso may omit some contextually important variables that are weakly predictive in the statistical sense, its selections still highlight the core confounding pathways. This outcome suggests that our original model specification was comprehensive, and that the main drivers of behavioral responses to the flavor ban were concentrated in a relatively small set of key covariates.

6 Discussion

This study set out to evaluate the impact of state-level e-cigarette flavor bans on vaping and other substance use behaviors using BRFSS data from 2016 to 2023. While the main findings show significant reductions in vaping and possible spillovers into traditional tobacco products, several caveats, alternative interpretations, and modeling challenges must be discussed and examined.

One major finding of the paper is that flavor bans reduce current vaping behavior among individuals under 30. However, this effect appears to weaken as mental health deteriorates. Those reporting frequent poor mental health exhibit smaller reductions in vaping probability post-ban, and in some models, show no significant change at all. This suggests that individuals experiencing psychological distress may be more resistant to policy-induced behavioral change, particularly when nicotine use serves a coping or self-regulation function. While the logit model confirms effect heterogeneity across mental health subgroups, the OLS results

do not consistently detect it. The methodological divergence is in itself a meaningful result. One explanation is that logit captures marginal shifts more effectively in low-probability regions, while OLS averages over a wider probability distribution, diluting effect sizes. We try not to over-interpret these differences, but they highlight the importance of using both linear and nonlinear models in subgroup analysis.

In the context of spillovers, the results show that flavor bans may lead to a substitution effect toward combustible cigarettes and chewing tobacco. This is concerning from a public health standpoint, as these substances are known to carry higher health risks than ENDS. Yet, this substitution effect appears to be uneven across mental health status. individuals with frequent mental distress are less likely to increase smoking post-ban. A plausible explanation is that these individuals exhibit stickiness to ENDS and are unwilling or unable to switch to other forms of nicotine. This narrative complements our earlier finding that flavor bans are less effective among mentally vulnerable individuals in reducing vaping. Taken together, these findings suggest that flavor bans may not only be less effective for high-risk subgroups but may also shift behavior in unintended ways for the general population.

While our difference-in-differences framework yields intuitive results, the event-study estimates offer a more complex picture. Figure 3(a) supports the validity of the parallel trends assumption for vaping, but the post-treatment coefficients show a positive rather than negative shift in usage, contrary to our main specification. Figure 3(b), which captures smoking behavior, presents noisy pre-trends and post-treatment coefficients that move in the opposite direction of the hypothesized substitution. One likely explanation for these discrepancies is data sparsity. Our sample includes a limited set of states with implemented flavor bans and only a few years of data on either side of the policy. This limited time variation weakens the reliability of the event-study estimator, which depends on stable pre-treatment trends and sufficient post-treatment observations. Additionally, the staggered timing of bans across states may introduce complications that standard event-study designs cannot fully address without correction (e.g., via Gardner-style stacked estimators).

A central limitation of the study is that it relies entirely on BRFSS data, which lacks flavor-specific information for ENDS products. This limitation is mentioned in Cotti et al. (2024) and restricts our analysis to subtle changes in vaping behavior without being able to disentangle flavored from unflavored product usage. This is especially relevant since many users may have switched to tobacco-flavored ENDS, which remain legal under most bans. Another limitation is our inability to access Youth Risk Behavior Survey (YRBS) state-level data, which requires special request and authorization. Since youth are disproportionately affected by these bans, the absence of youth-specific data prevents us from fully capturing the most impacted demographic. Furthermore, while our sample restricts to under-30 adults to partially address this issue, there remains a gap in understanding true youth substitution dynamics.

We also did not have the opportunity to implement the Gardner two-step difference-in-differences estimator. This method is known for addressing bias in staggered treatment timing designs and could have improved the precision and causal interpretability of our event-study results. Unfortunately, due to time and resource constraints, we were unable to estimate this model, which would have been a particularly valuable addition to strengthen our robustness checks.

Another potential issue is that state-level controls such as tax policies or indoor ENDS bans may be endogenous to vaping trends. Although we include these policies as controls to avoid omitted variable bias, their own implementation may have been triggered by the increase in ENDS use. In that case, the coefficient estimates may partially reflect reverse causality. Similarly, we are limited by the granularity of state-level data and cannot control for local enforcement differences, availability of unregulated products, or black market adaptations that may occur after formal restrictions.

Finally, while our use of double selection Lasso helps mitigate overfitting, it is not immune to excluding substantively important covariates. For instance, many of the state fixed effects and pandemic-period controls were not selected by the Lasso step, but may still hold

contextual significance. Nevertheless, the fact that the treatment effect estimate remains consistent even under this sparse specification suggests that our findings are not driven by model noise, reinforcing the robustness of our main conclusions.

7 Conclusion

This study evaluates the impact of e-cigarette flavor bans on current vaping and other substance use behaviors among individuals under 30, using BRFSS data from 2016 to 2023. Consistent with prior work, we find that flavor bans are associated with reduced vaping rates. However, these reductions are not uniform across the population. Individuals experiencing frequent mental distress appear less responsive to the ban, indicating behavioral inertia or dependence. In contrast, among the general population, flavor bans may trigger substitution into more harmful nicotine products such as cigarettes and chewing tobacco.

The findings suggest that while flavor bans may be effective in reducing ENDS use overall, they also carry risks of behavioral substitution and differential effects among vulnerable groups. Our subgroup analyses highlight the need for more nuanced tobacco control policies that account for mental health disparities. Though the event-study results raise questions about short-term causal dynamics, our main DiD results are robust across OLS, logit, and double Lasso specifications. The limitations of the study, particularly the absence of youth-specific data, a short time window, and the inability to implement advanced DiD estimators, point to directions for future work.

Taken together, this paper adds to the growing body of literature suggesting that one-size-fits-all public health regulations may not address the complexities of behavioral addiction and mental health. As nicotine regulation continues to evolve, careful consideration must be given to both intended and unintended consequences, especially for high-risk subgroups who may require more tailored interventions.

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Appendix

	count	mean	std	min	25%	50%	75%	max	unique
ECIGNOW2	196967	2.88	1.31	1	1	4	4	4	4
_RFSMOK3	195959	0.15	0.36	0	0	0	0	1	2
_RFBING6	190494	0.31	0.46	0	0	0	1	1	2
USENOW3	196777	0.06	0.23	0	0	0	0	1	2
MARIJAN1	196967	0.82	0.38	0	1	1	1	1	2
MENTHLTH	193870	6.93	9.34	0	0	3	10	30	31
unemp	195234	4.15	1.74	1.7	3	3.8	4.7	30.5	139
Death	196967	281.74	673.8	-3488	0	16	283	18664	832
tax	189876	1.97	1.18	0.17	1.03	1.7	3.02	4.35	55
_AGE80	196967	23.73	3.4	18	21	24	27	29	12
SEXVAR	196967	1.44	0.5	1	1	1	2	2	2
_RACE	193923	2.83	2.78	1	1	1	4	8	8

Table A1: Summary statistics of key variables in the analysis sample of individuals under 30.

Variable	**Coefficient**	**Std. Error**	**P-Value**
Flavor Ban	0.0117	0.006	0.036
Tobacco 21	-0.0166	0.004	0.000
Combustible Cigarette Tax	0.0076	0.003	0.003
ENDS Tax	-0.0057	0.015	0.707
COVID Dummy	-0.0252	0.016	0.124
Recreational Marijuana Legal	0.0010	0.003	0.759
Post Indoor ENDS Ban	-0.0075	0.006	0.222
Age	0.0011	0.000	0.000
Poor Mental Health Days	0.00008	0.00009	0.386
Unemployment Rate	0.0006	0.001	0.252
Observations	183,864		
Model Type	WLS with Clustered SEs		

Table A2: OLS regression estimating the effect of flavor bans on current chewing tobacco use among individuals under 30. Controls include demographic and policy covariates, state and year fixed effects, and BRFSS analytic weights. Robust standard errors clustered at the state level.

Variable	**Coefficient**	**Std. Error**	**P-Value**
Flavor Ban	0.0280	0.010	0.004
Tobacco 21	-0.0438	0.015	0.003
Combustible Cigarette Tax	0.0101	0.008	0.204
ENDS Tax	-0.0350	0.039	0.371
COVID Dummy	-0.0679	0.021	0.001
Recreational Marijuana Legal	0.0106	0.007	0.153
Post Indoor ENDS Ban	-0.0148	0.014	0.295
Age	0.0150	0.001	0.000
Poor Mental Health Days	0.0042	0.000	0.000
Unemployment Rate	0.0030	0.002	0.063
Observations	183,134		
Model Type	WLS with Clustered SEs		

Table A3: OLS regression estimating the effect of flavor bans on current cigarette use among individuals under 30. Controls include demographic and policy covariates, state and year fixed effects, and BRFSS analytic weights. Robust standard errors clustered at the state level.

Variable	**Coefficient**	**Std. Error**	**P-Value**
Flavor Ban	-0.0156	0.004	0.000
Moderate Mental Distress (1–9 days)	0.0066	0.002	0.000
Frequent Mental Distress (10+ days)	0.0657	0.002	0.000
Flavor Ban × Moderate MH	-0.0019	0.004	0.641
Flavor Ban × Frequent MH	0.0013	0.004	0.766
Tobacco 21	0.0232	0.006	0.000
COVID Dummy	0.0224	0.010	0.025
Post Indoor ENDS Ban	0.0125	0.004	0.001
Poor Mental Health Days	—	—	—
Age (Linear)	-0.0020	0.000	0.000
Unemployment Rate	-0.0011	0.001	0.101
Observations	184,026		
Model Type	WLS with clustered SEs		

Table A4: Linear probability model estimating flavor ban effects on current vaping by mental health subgroup, using interaction terms. No statistically significant interaction effects are found, though signs are consistent with logit estimates shown in the main text.

Variable	**Coefficient (Lasso)**
MENTHLTH	0.288
SEXVAR_2	-0.101
_RACE_2.0	-0.086
_RACE_8.0	-0.074
_AGE80	-0.052
survey_year_2020	0.014
_RACE_4.0	-0.013
tax	-0.001

Table A5: Covariates selected via post-double selection Lasso procedure. Variables retained include mental health, age, sex, race indicators, and select year/policy variables. The table reports non-zero coefficients from the Lasso stage used in final model estimation.